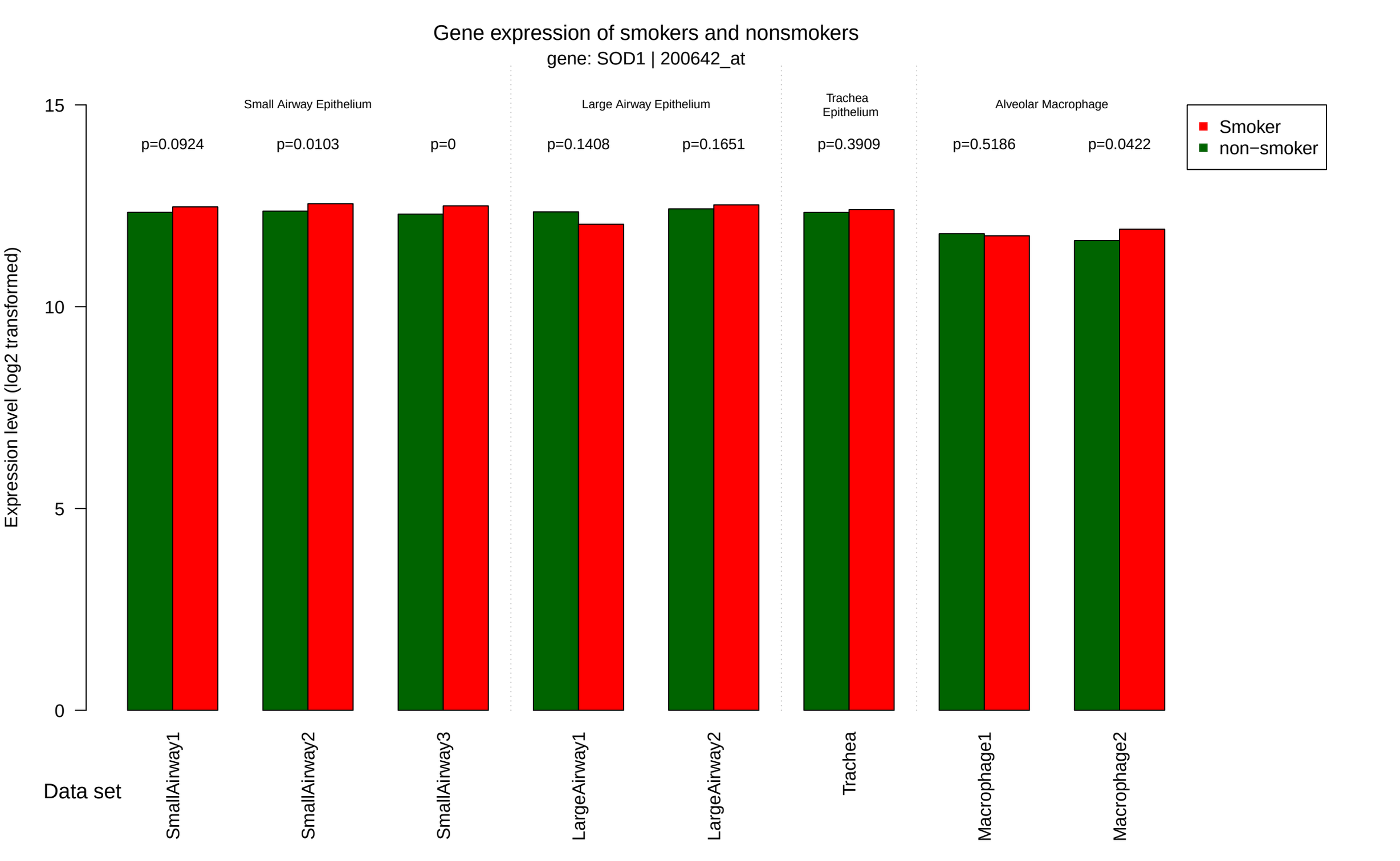
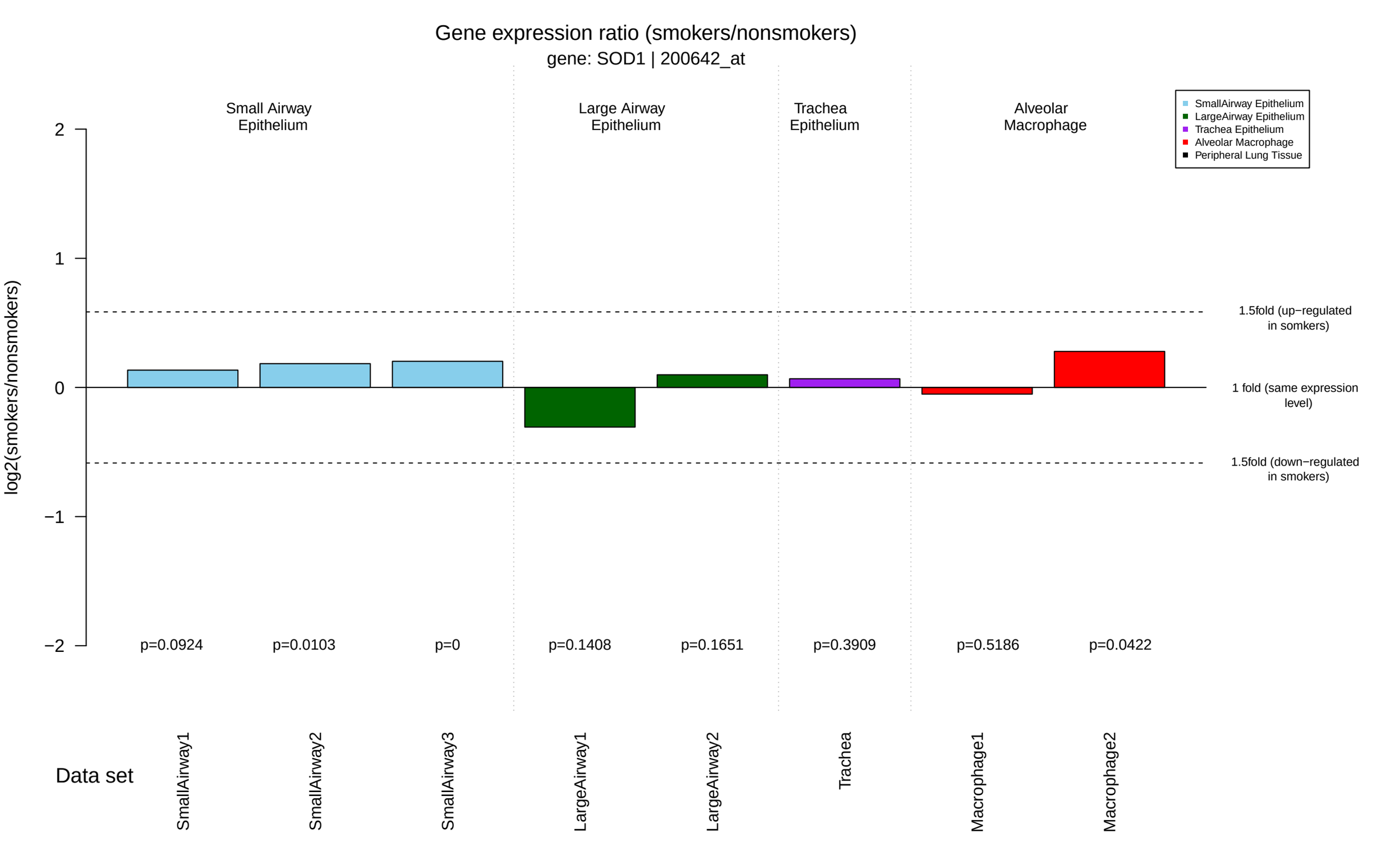
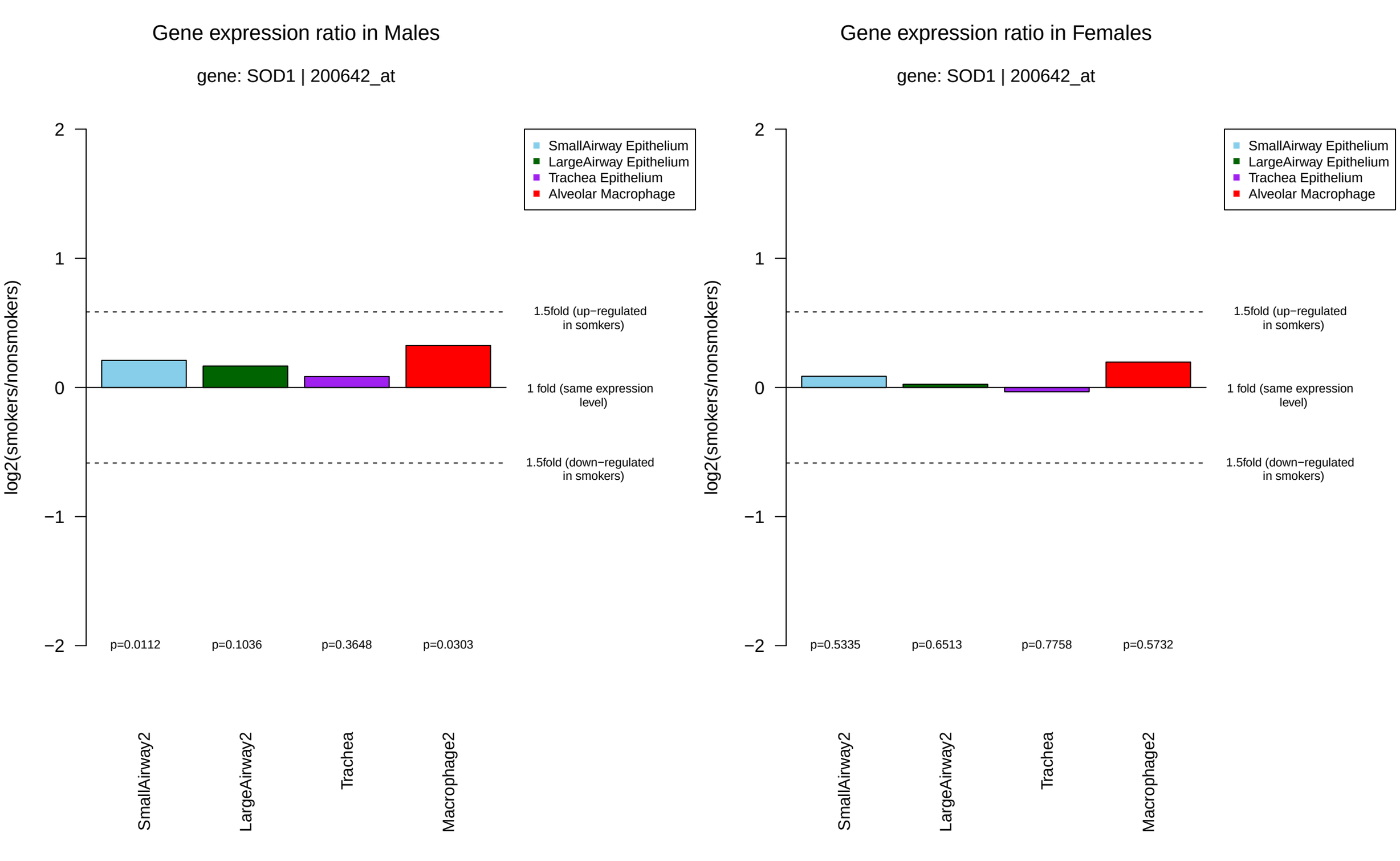
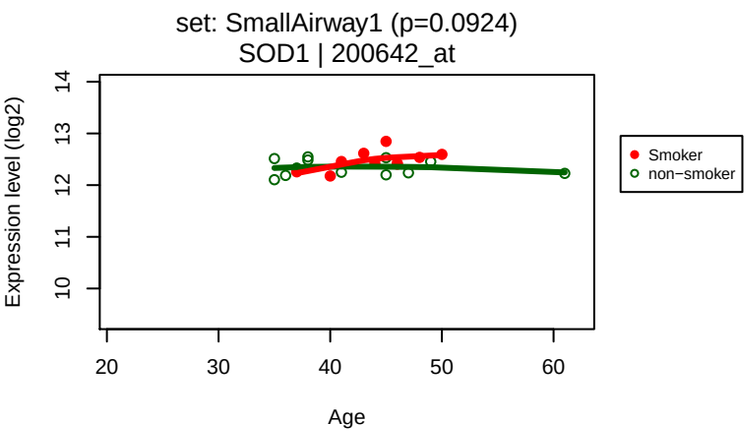
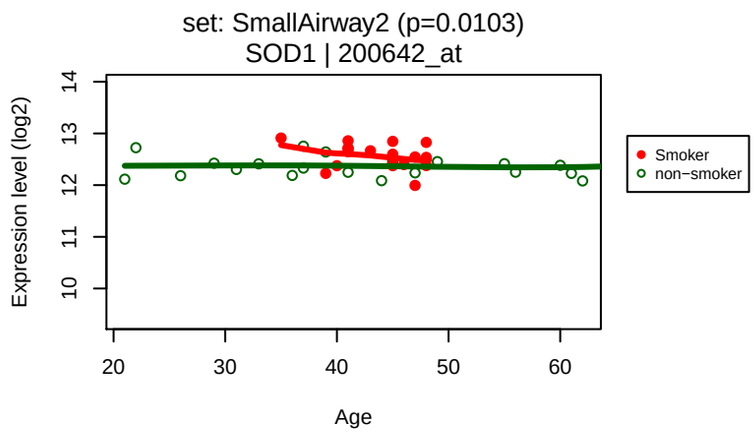
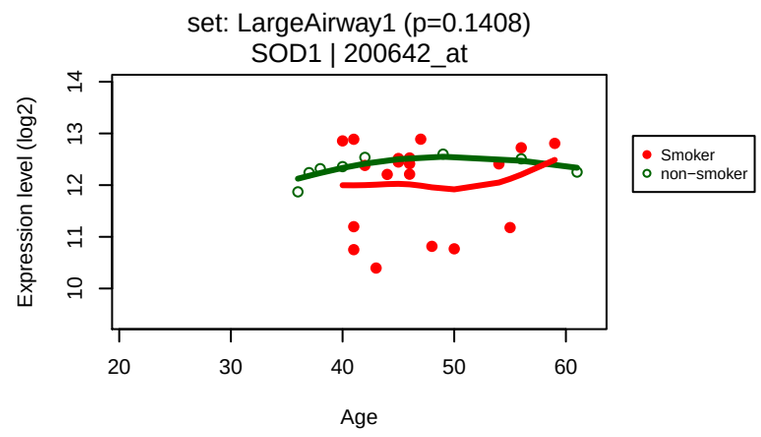

Gene expression of smokers and nonsmokers
gene: SOD1 | 200642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0924
p=0.0103
p=0
p=0.1408
p=0.1651
p=0.3909
p=0.5186
p=0.0422
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

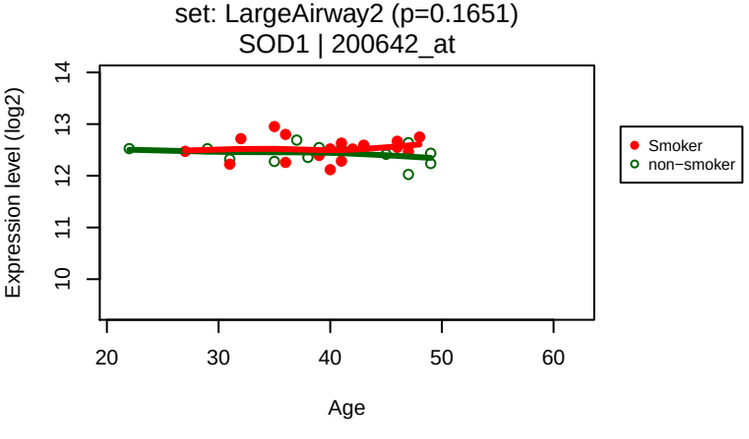

Gene expression ratio (smokers/nonsmokers)
gene: SOD1 | 200642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0924
p=0.0103
p=0
p=0.1408
p=0.1651
p=0.3909
p=0.5186
p=0.0422
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

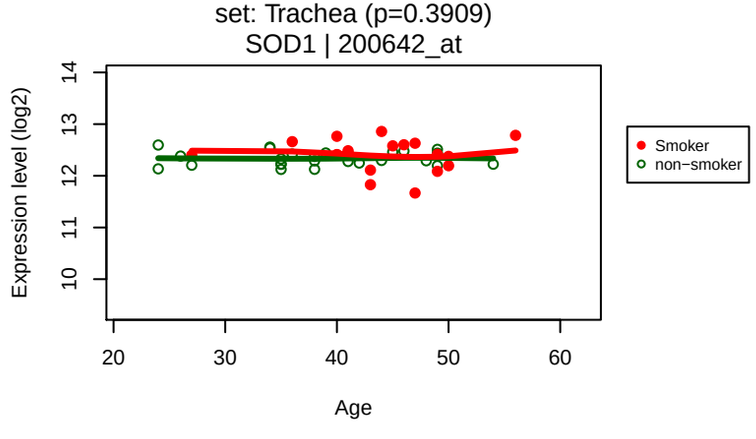

Gene expression ratio in Males
gene: SOD1 | 200642_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0112
p=0.1036
p=0.3648
p=0.0303
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: SOD1 | 200642_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5335
p=0.6513
p=0.7758
p=0.5732
SmallAirway2
LargeAirway2
Trachea
Macrophage2

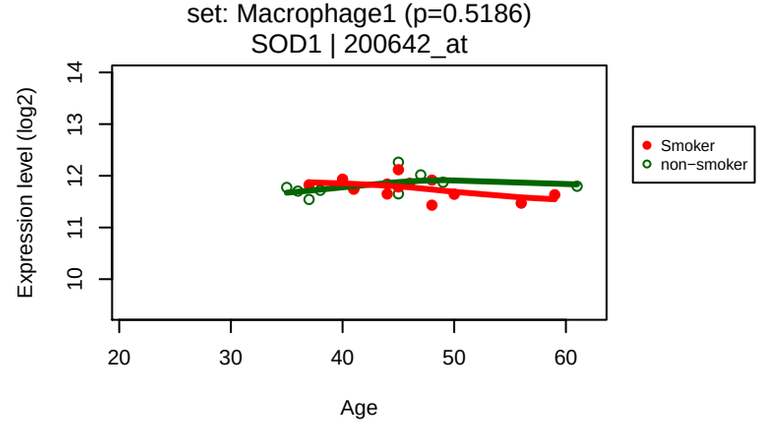

set: SmallAirway1 (p=0.0924)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker

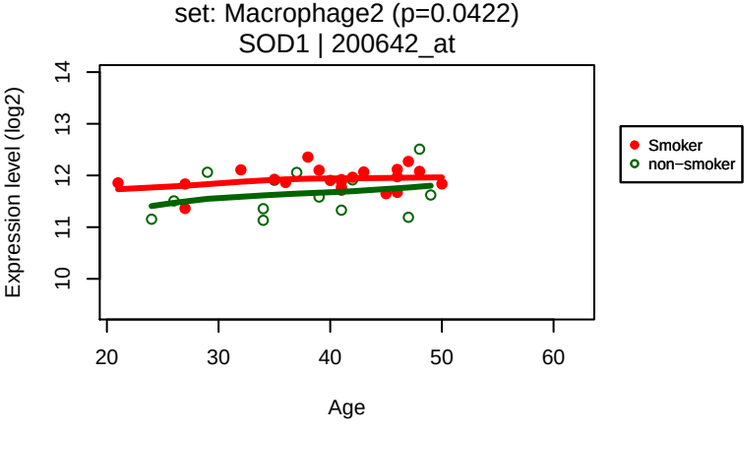

set: SmallAirway2 (p=0.0103)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker

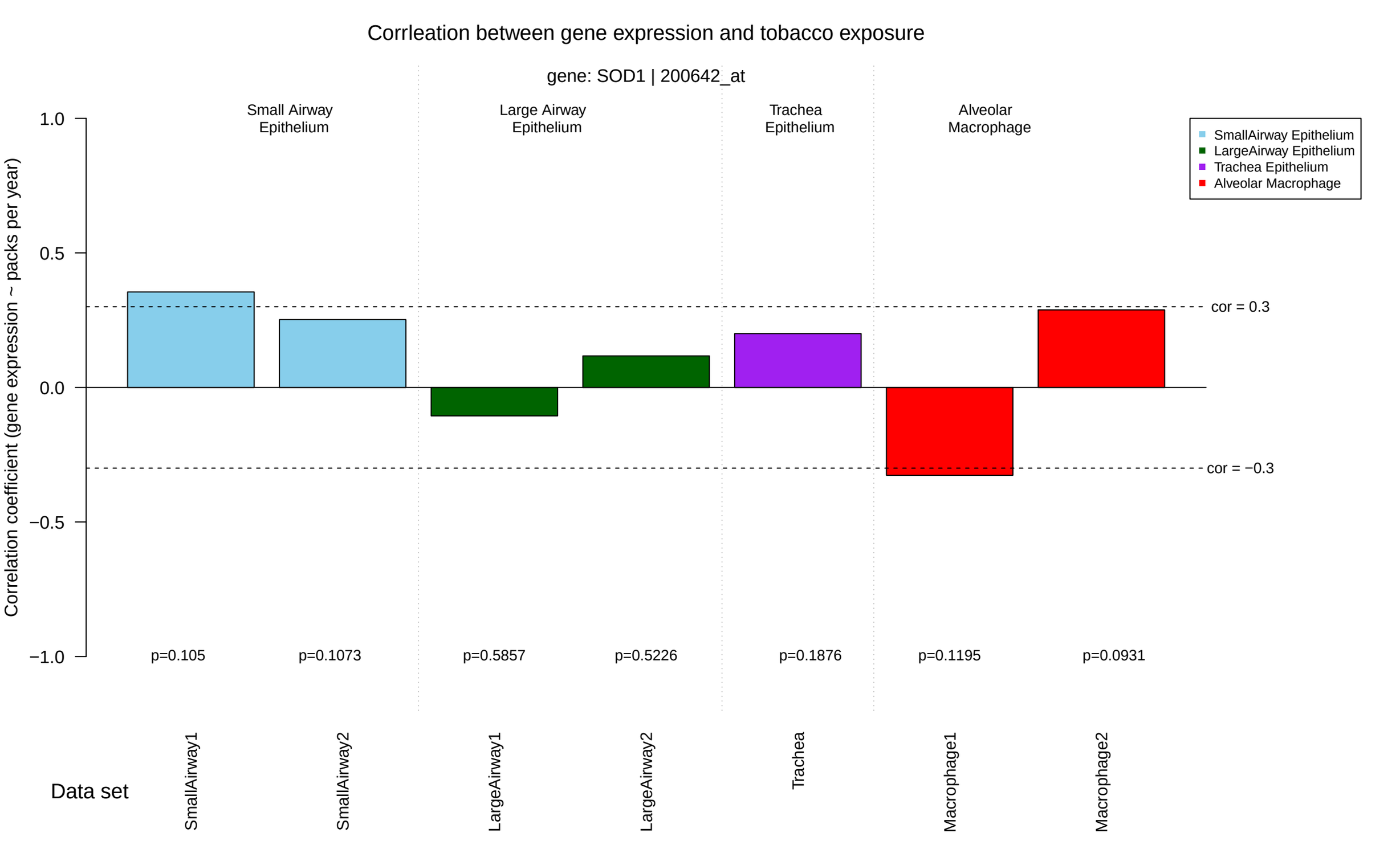

set: LargeAirway1 (p=0.1408)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1651)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3909)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5186)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0422)
SOD1 | 200642_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: SOD1 | 200642_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.105
p=0.1073
p=0.5857
p=0.5226
p=0.1876
p=0.1195
p=0.0931
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2